Spencer Shirah

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EDUCATION

Auburn University, Auburn, AL
Bachelor of Science in Software Engineering — Expected Summer 2026

EXPERIENCE

Software Consultant / Founder – Shackmont Software (March 2026 – Present)

- Designed and developed custom software solutions for local businesses, including multi-page websites and interactive tools
- Built a roofing cost estimator integrating external APIs to generate real-time pricing for customers
- Delivered production-ready web applications focused on performance, usability, and business impact
- Managed client requirements, deployment, and iterative improvements independently

ENGINEERING PROJECTS

Keyword Extraction & Evaluation Pipeline (**Python**, LLMs, BERT, NLP, Data Processing)

- Engineered a multi-stage pipeline to extract structured keywords from unstructured, noisy text data
- Implemented a hybrid approach combining LLM-based generation and BERT-based similarity scoring
- Designed an agent-based feedback loop to iteratively refine outputs and reduce hallucinations
- Developed parallel extraction workflows to handle variable-length inputs and edge cases
- Built evaluation framework using relevance scoring and human-aligned criteria
- Processed datasets with configurable batching to improve scalability and performance

Liver Biopsy Simulator (JavaScript, **Three.js**, Simulation Systems)

- Developed an interactive simulation modeling liver biopsy procedures for training and decision support
- Implemented 3D visualization using Three.js to represent needle positioning and procedural interaction
- Designed state-based workflows to simulate varying conditions and edge cases
- Focused on accuracy, reliability, and user interaction in a precision-driven environment

AI Phone Receptionist System (**Python**, APIs, Backend Systems, Twilio, OpenAI)

- Built a real-time voice-processing system to handle inbound calls and route requests based on classified intent
- Designed API-driven architecture integrating speech input, NLP processing, and response handling
- Implemented fault-tolerant conversational flows to ensure reliability under unpredictable user input

Secure Password Manager (**Python**, Cryptography, Security)

- Developed an encrypted credential storage system using modern cryptographic techniques
- Implemented secure key derivation and encryption to protect sensitive data
- Designed system architecture aligned with real-world security and threat modeling principles

Personal Portfolio Website (HTML, CSS, **JavaScript**, GitHub)

- Designed and deployed a responsive portfolio showcasing engineering and cybersecurity projects
- Implemented interactive UI elements and contact functionality for recruiting engagement
- Managed version control and deployment using Git and GitHub

LEADERSHIP & EXPERIENCE

Ethical Hacking Club, Auburn University (Member | Sept 2025 – Present)

- Conducted penetration testing labs including SQL injection, buffer overflows, and network exploitation
- Collaborated to identify and remediate system vulnerabilities in controlled environments

Chi Phi Fraternity, Auburn University (Member | Sept 2024 – Present)

- Completed 50+ hours of community service and participated in fundraising initiatives
- Coordinated events and collaborated across teams to support philanthropic efforts

TECHNICAL SKILLS

Languages: Python, C, C++, Java, SQL, JavaScript
Systems & Tools: Git, GitHub, Linux, APIs, MySQL, Milvus
Technologies: OpenAI API, Twilio, Web Development (HTML/CSS)